4.5 Detecting ranges using logical operators

Settings

Beta features

logical operator

A logical operator treats operands as being True or False, and eval
Logical operators include AND, OR, and NOT.

PARTICIPATION ACTIVITY 4.5.1: Logical operators: AND, OR, and NOT.

Start 2x speed

a	b	a AND b
False	False	False
False	True	False
True	False	False
True	True	True

a	b	a OR b
False	False	False
False	True	True
True	False	True
True	True	True

a
False
True

Let x = 7, y = 9

(x > 0) AND (y < 10) True
True True

(x < 0) OR (y > 10) False
False False

NOT (x <
False

(x > 0) AND (y < 5) False
True False

(x < 0) OR (y > 5) True
False True

NOT (x >
True

Captions

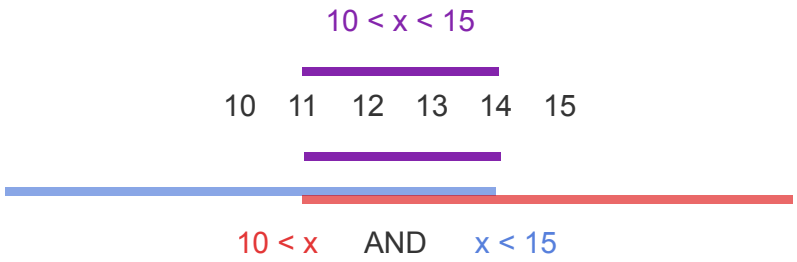
1. AND evaluates to True only if BOTH operands are True.
2. OR evaluates to True if ANY operand is True (one, the other, or both).
3. NOT evaluates to the opposite of the operand.
4. Each operand is an expression. If x = 7, y = 9, then (x > 0) AND (y < 10) is True AND True, so True (both operands are True).

Table 4.5.1: Logical operators.

Logical operator	Description
a AND b	***Logical AND***: True when both of its operands are Tru
a OR b	***Logical OR***: True when at least one of its two operan
NOT a	***Logical NOT***: True when its one operand is False, and

PARTICIPATION ACTIVITY 4.5.3: Using AND to detect if a value is within a range.

Start 2x speed



Captions

1. The range 10 < x < 15 means that x may be 11, 12, 13, or 14.
2. Specifying that range in a program can be done using two < operators along with an AND o and higher.
3. x < 15 defines the range 14 and lower. ANDing yields the overlapping range. Only when x is expressions be true.

Boolean

A Boolean refers to a value that is either True or False.

Figure 4.5.1: Creating a Boolean.

```
my_bool = True   # Assigns my_bool with the boolean value True

is_small = 4 < 3  # Assigns is_small with the result of the expression 4 < 3 (F
```

and / or / not

Keywords and, or, and not (lowercase) are used to represent the A operators.

Table 4.5.2: Logical operators.

Logical operator	Description
a **and** b	***Boolean AND***: True when both operands are True.
a **or** b	***Boolean OR***: True when at least one operand is True.
not a	***Boolean NOT*** (opposite): True when the single operand is False (and

Table 4.5.3: Logical operators examples.

Given age = 19, days = 7, user_char = "q"

`(age > 16) and (age < 25)`	True, because both operands are True.
`(age > 16) and (days > 10)`	False, because both operands are not Tru
`(age > 16) or (days > 10)`	True, because at least one operand is Tru
`not (days > 10)`	True, because operand is False.
`not (age > 16)`	False, because operand is True.
`not (user_char == "q")`	False, because operand is True.

Figure 4.5.2: Detecting ranges: Cable TV channels.

```
if (user_channel >= 2) and (user_channel <= 499):
   channel_type = "s"

elif (user_channel >= 1002) and (user_channel <= 1499):
   channel_type = "h"

else:
   channel_type = "e"
```

Try 4.5.1: Detecting ranges: Cable TV channels.

Run the program, enter 3 as input, and observe the output. Run the program again with another r

PARTICIPATION ACTIVITY 4.5.8: Detecting ranges implicitly vs. explicitly.

Start 2x speed

```
if x < 0 :
    # Negative
```

```
if x < 0 :
  # Negative
```

```
elif  (x >= 0) and (x <= 10) :
    # 0..10

elif  (x >= 11) and (x <= 20) :
    # 11..20

else :
    # 21+
```

Explictly defined ranges

```
elif (x <= 10) :
  # 0..10

elif (x <= 20) :
  # 11..20

else :
  # 21+
```

Implictly defined rang

Captions

1. This code detects ranges explicitly using the AND operator. The first branch executes when (x >=0) and (x <= 10).
2. If the first branch doesn't execute, x must be >= 0. So the second branch's expression is x <
3. Implicit ranges can simplify a multi-branch if the statement for ranges is without gaps.

CHALLENGE ACTIVITY 4.5.1: Detecting ranges using logical operators.

763650.1722066.qx3zqy7

Start

Modify the given if statement so that 'Large town' is output if num_residents is in the range 11(

Click here for examples

```
num_residents = int(input())

# Modify the following line
if (num_residents >= 1100) or (num_residents <= 4600):
    print("Large town")
else:
    print("Not a large town")
```

1 2 3

Check Next level

View solution *(Instructors only)*